

Tutorial 8 — Exercise 1: Deriving the F -Test

Introductory Econometrics — TA Session

Duration: 50 minutes | Based on Lecture 14 (Marmer)

Consider the log-wage equation from Lecture 14:

$$\ln(Wage_i) = \beta_0 + \beta_1 Exp_i + \beta_2 Exp_i^2 + \beta_3 PrevExp_i + \beta_4 PrevExp_i^2 + \beta_5 Educ_i + U_i$$

where Exp is current job experience, $PrevExp$ is previous experience, and $Educ$ is years of education ($k = 5$ regressors plus intercept). We want to test whether previous experience has *any* effect on wages. This requires testing β_3 and β_4 jointly. You will derive the test statistic that does this correctly.

Step 1. Hypotheses. The null imposes $q = 2$ restrictions simultaneously:

$$H_0: \beta_3 = 0 \text{ and } \beta_4 = 0 \quad \text{against} \quad H_1: \beta_3 \neq 0 \text{ or } \beta_4 \neq 0.$$

(a) Suppose H_0 is true. You run two separate size- α t -tests and reject whenever *at least one* rejects. Using $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, show that the false rejection probability exceeds α .

Step 2. Unrestricted and restricted models. We compare the fit of two models:

- **Unrestricted:** all regressors included $\rightarrow$ compute $SSR_{ur} = \sum \hat{U}_i^2$. Has $k + 1 = 6$ parameters; residual df $= n - k - 1$.
- **Restricted** (impose H_0 : drop $PrevExp$, $PrevExp^2$): $\ln(Wage_i) = \beta_0 + \beta_1 Exp_i + \beta_2 Exp_i^2 + \beta_5 Educ_i + U_i$ $\rightarrow$ compute SSR_r . Has 4 parameters.

(b) Verify that the parameter difference equals q . Explain why $SSR_r \geq SSR_{ur}$ always.

Step 3. The F -statistic. If H_0 is true, $SSR_r - SSR_{ur}$ should be small (restrictions barely hurt fit). If false, it should be large. The F -statistic formalises this:

$$F = \frac{(SSR_r - SSR_{ur}) / q}{SSR_{ur} / (n - k - 1)}$$

(c) Interpret the numerator and denominator. Why divide by q ? What does $SSR_{ur} / (n - k - 1)$ estimate?

(d) Compute F using Lecture 14 data: $SSR_{ur} = 96.998$, $SSR_r = 99.463$, $n = 526$, $k = 5$, $q = 2$.

Step 4. Distribution and degrees of freedom. Under H_0 and normality ($U_i|X \sim \mathcal{N}(0, \sigma^2)$): $F \sim F_{q, n-k-1}$. This follows because $(SSR_r - SSR_{ur}) / \sigma^2 \sim \chi_q^2$ and $SSR_{ur} / \sigma^2 \sim \chi_{n-k-1}^2$, independently, and the F -distribution is defined as the ratio of two independent χ^2 's divided by their df (with σ^2 cancelling).

(e) State the numerator and denominator df for the wage example. Explain why $F \geq 0$ always.

Step 5. Decision rule. At significance level α : reject H_0 when $F > F_{q, n-k-1, 1-\alpha}$.

The F -test is always one-sided (reject for large F only) and the denominator df comes from the **unrestricted** model.

(f) The critical value is $F_{2, 520, 0.95} \approx 3.01$ and the p -value ≈ 0.0015 . Do you reject? Interpret.

Step 6. R^2 version. When both models share the same dependent variable ($SST_r = SST_{ur} = SST$):

(g) Using $SSR = SST(1 - R^2)$, show that $F = \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n - k - 1)}$. Verify with $R_{ur}^2 = 0.3461$, $R_r^2 = 0.3294$.